

Assignment 1 — Patient Clinical Data Analysis

Dataset

patient_clinical_data_raw.csv

Objective: You are working as a data analyst in a hospital. The hospital has provided a raw patient dataset containing demographic, clinical, admission, and treatment information. Your task is to use **Pandas** to clean, transform, analyze, and summarize the data.

Part A — Understanding the Dataset

1. Load the dataset into a Pandas DataFrame.
2. Display the first 10 and last 10 records.
3. Find the number of rows and columns.
4. Display the column names and their data types.
5. Generate a statistical summary of the numerical columns.
6. Identify the number of unique values in each column.

Part B — Data Cleaning

7. Identify all columns containing missing values.
8. Calculate the percentage of missing values in each column.
9. Handle missing values in the numerical columns appropriately.
10. Identify and remove duplicate records.
11. Check the Age column for invalid values.
12. Correct or handle patients whose age is outside a reasonable range.
13. Identify inconsistent values in categorical columns such as Gender and Department.
14. Standardize these categorical values.

Part C — Data Transformation

15. Create a new column called Age_Group:

< 18	Child
18–40	Young Adult
41–60	Middle Age
> 60	Senior

16. Separate Blood_Pressure into two columns:
 - Systolic
 - Diastolic
17. Create a High_Risk column based on suitable clinical conditions.
18. Create a Cost_Per_Day column using treatment cost and length of stay.

Part D — Analysis

19. Calculate the average age of patients.
20. Find the number of patients in each department.
21. Calculate the average treatment cost for each department.
22. Find the average length of stay for each diagnosis.
23. Identify the top 10 patients with the highest treatment cost.
24. Find the department with the highest average treatment cost.
25. Calculate the percentage of patients who were readmitted.
26. Compare readmission rates across departments.
27. Find the average cholesterol and glucose level by diagnosis.
28. Identify patients with:
 - high glucose
 - high cholesterol
 - high heart rate